8th Dec, 2025

IRS :-

IRS - Information Retrieval system.

Data → Raw information.

Ex:- Alphabets, numbers, Images, Audio, video, Binary values.

Information = Processed data.

Ex:- sum = a+b.

Here a, b are data and it is being processed so sum is information

Data is three types :-

① structural	② semistructured	③ unstructured.
- well organised data	- structured + unstructured	- No structured (fixed format)
- stored in table	Ex:- Documents, HTML pages, XML data etc.	- No rows/columns
- rows/columns (not there in)	- To get this data we use Data ware house.	- Hard to store table
- easy to search & process		- very common in real life.
- single row is called record/tuple.		- single row is called item.
Ex:- sql, mysql, oracle		Ex:- Text msgs, HTML, XML (pages), Image data, newspapers.
- To get this data we use DBMS.		- To get this data we use IRS.
- When we search through DB then we get exact info.		- When we search through IRS then we get relevant data/info.

Definition :-

IRS :- IRS is a computer system that helps users find useful information from large collection of data.

Ex:- google search, youtube, library search information.

→ Info searched can be Images, Audio, video, newspaper, multimedia objects, Numeric data.

- Non-numeric data.

Item :-

- Single row in a unstructured data.
- Small complete unit → Information.

Ex:- Item ⇒ C_2 Class total information.

Information :-

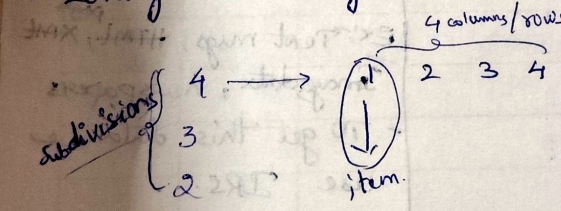
- Hardware or s/w application.
- To get any info we need device (like pc, phone, desktop) hardware, memory, storage, google search (application to search) but to search network b/w them is also important.

Ex:- Data → google search.

Images : WEBSEEK, BITTO.Com, ALTAVISTA/IMAGES.

Disney : Video indexing

→ zoning → making subdivisions. Can make 4 or 3 or 2 subdivisions



→ we need some components to store data.

Basic :- (online)

* hardware
* s/w
* storage
* memory
* information
* application.

phone, Desktop, PC's.

In offline :-

* h/w
* s/w
* pdf

objectives of IR

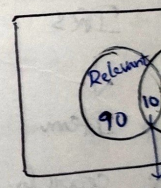
* It reduces ove
* It first calcula and recall of s
* precision ⇒ chec
  recall ⇒ check
* Needed Inform
  - Relevant Inf
  - Non Relevar
* Formulas :-

Precision = |{Relev

R

Recall = |{Relevant

Ex:-



Formula :- n(A∩B

precision =

Recall = 10/100

Precision → Di
non-r

Recall → not eff

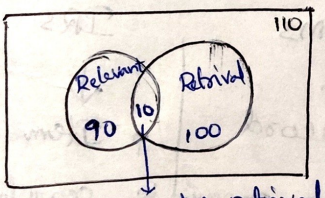
objectives of IRS :-

* It reduces overhead & burden.
* It first calculate the precision and recall of system.
* precision ⇒ checks Quantity
  recall ⇒ checks Quality.
* Needed Information : 2 Types.
  - Relevant Info
  - Non Relevant Info.
* Formulas :-

$$\text{Precision} = \frac{|\{\text{Relevant}\} \cap \{\text{Retrieved}\}|}{\text{Retrieved}}$$

$$\text{Recall} = \frac{|\{\text{Relevant}\} \cap \{\text{Retrival}\}|}{\text{Relevant}}$$

Ex:-



Relevant & Retrival

n(A) = 90
n(B) = 100

10 just taken randomly

Formula :- $n(A \cap B) = n(A) + n(B) - n(A \cup B)$

$= 90 + 100 - 180$
$= 10$?

$$\text{Precision} = \frac{10}{110} \rightarrow (100+10)$$

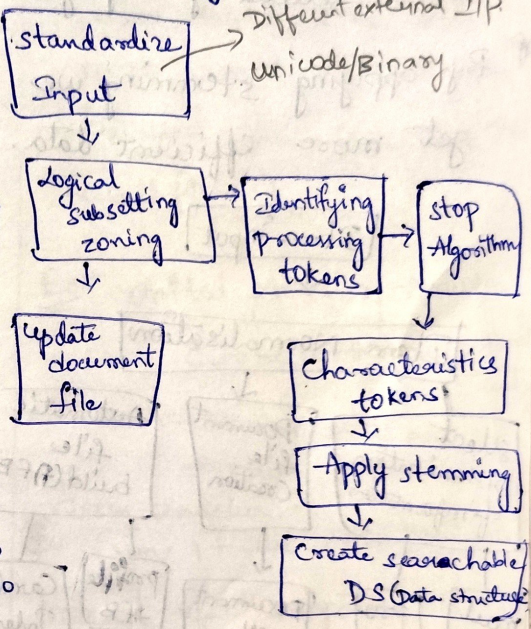
$$\text{Recall} = \frac{10}{100} \rightarrow (90+10)$$

Precision → directly effected by non-relevant data.

Recall → not effectly by non-relevant data.

Functional Overviews :-

* ① Item Normalisation.
  ↓
  small complete unit.
  - (i) Integrated → we first Normalise & language encoding (or) fileformat.
  - (ii) standard format
  - (iii) zoning
  - (iv) Indexing



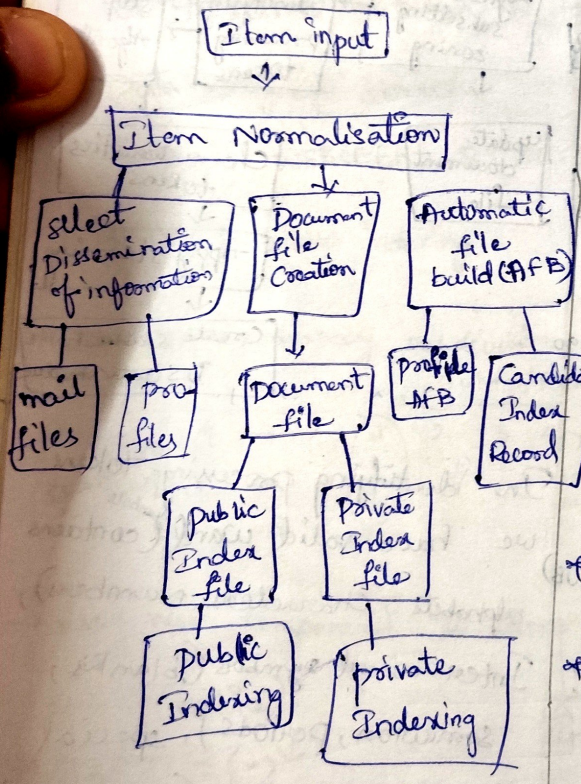
In 'identifying processing tokens we have valid word symbols (contains alphabets, characters, numbers), Inter-word symbols (blanks, semicolon, periods), special processing symbols (- , hyphen).

* we can't use special processing symbols alone. we use them in combination of valid word symbols & sometimes Inter-word.

* In stop algorithm, it selects and give only our mostly searched tokens and removes least searched tokens (which means they are irrelevant).

* Characteristics tokens shows relevant data of each searched tokens selected by stop algorithm.

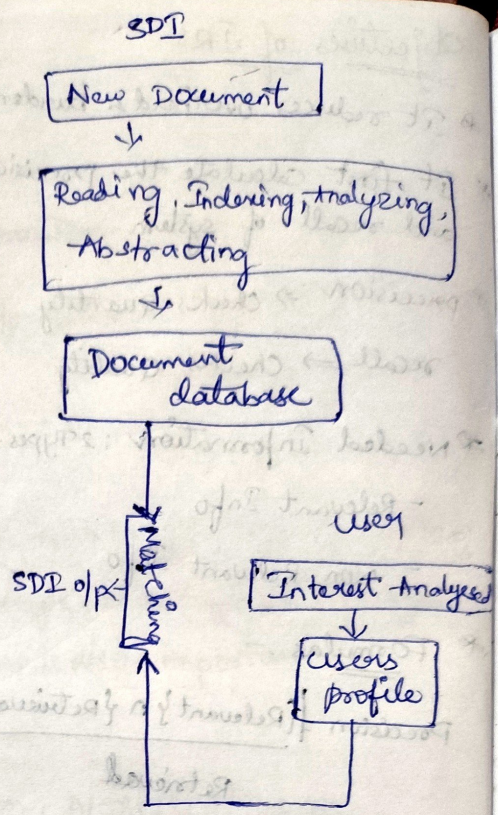
* By applying stemming we get more efficient data.



SDI :-

* Selective dissemination of Info.

* It do distribution spreading



Relationship to DBMS :-

DBMS	IRS
* Record	* Item
* Query processing	* search, browsing
* we get exact data	* Relevant data
* get structured data	* Unstructured data

* To get both features of IRS and DBMS. We will combine both like :-

IRS + DBMS → Query process → get exact data (or) most relevant

Ex:- For IRS + in now INQUIRE → It's been

IRS capa

① search
② Browse
③ Mis cell
④ standa

① Search cap

We can sear ways :- off

& Types :- → fu

① Boolean l
② proximi
③ Contigu
④ Fuzzy s
⑤ Term
⑥ Numeri
⑦ NLQ
⑧ Concept

* Boolean lo NOT. we set union, set Inte

Ex:- For IRS+DBMS we have in nowadays:-

INQUIRE DBMS

↓

It's been 15 years.

IRS capabilities:-

① search — offline, online
② Browse
③ Miscellaneous
④ standard.

① Search capabilities:-

We can search in two ways:- offline and online.

8 Types:- → functions

① Boolean logic
② proximity
③ Contiguous word
④ Fuzzy searches
⑤ Term masking
⑥ Numeric/Data Range
⑦ NLQ
⑧ Concept/thesaurus

* ① Boolean logic means - AND, OR, NOT. we implement using set union, set difference and set Intersection.

* ② In proximity, only 2 words are taken and only 1 adjacent (AND, OR, NOT) is used.

Ex:- Compute AND Design

It is taken from left to right.

③ In contiguous word, more than 2 words and more than 1 adjacents can be used. Here precision ↑es, Recall (↑low).

Ex:- United states of America.

↓

Can be written as (optional, can be used)

* United ~~AND~~ states AND of America.

④ Fuzzy searches:-

Even if we write "computer" instead of "computer" it, automatically takes computer based on search history by IRS.

5. Term masking.

IRS automatically generates related content to the typed text.

Ex:- when we type reading it automatically gives data like stories, novels, news.

There are 2 types:- ① fixed length ② variable length

- Fixed length :- It again has 2. They are prefix and suffix of computer.

  Prefix : mini computer, mainframe computer.

  Suffix : computers, computing.

- Variable length.

  It has format * word *.

  Ex:- computer

  like => mainframe computers

(vi) Date / Numeric Range :-

Date Range has 2 types

- Variable length.
- Fixed length - distinct values.

  Ex:- 1 to 12 months, 1 to 31 days

→ Numeric Range has suffix, prefix.

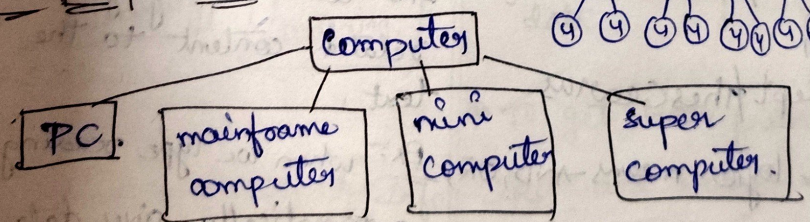
(vii) Concept / Thesaurus :-

→ They are in tree structure.

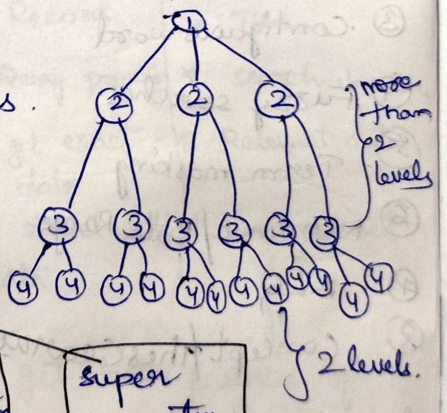
→ concept has more than 2 levels.

→ Thesaurus has 2 levels.

Example of Thesaurus :-



Ex. of concept



(viii) NLQ :- (Natural language query)

→ we always search with boolean logic

→ So, we search with NLQ with boolean logic

→ Everything in search under NLQ.

Browse capabilities :-

3 types of browse ca
i) Ranking ii) zoning

• The relevant data when searched is us "browsing"

(i) Ranking :-

→ After searching we data to browse in t ranking.

Browse the relevant data 1. 2 3

→ The ranking is based data, Downloads, feedbacks.

→ It also has visual and colouring.

Ex:- Using ratings colours to represent green - good.

(ii) zoning :-

The item is subdivi Sub-divided one i

→ It has range from

→ 1.0 means mostly

→ Everything in search comes under NLQ.

Browse capabilities:-

3 types of browse capabilities
i) Ranking ii) Zoning (iii) Highlighting

→ The relevant data obtained when searched is used for "browsing"

(i) Ranking :-

→ After searching we get relevant data to browse in the form of ranking.

search
Browse the relevant data
1. ---
2. ---
3. ---

→ The ranking is based on frequently searched data, Downloads, views, feedbacks.

→ It also has visual representation and colouring.

Ex:- Using ratings, using colours to represent like red=bad, green - good.

(ii) Zoning :-

The item is subdivided & each sub-divided one is item.

→ It has range from 0.0 - 1.0

→ 1.0 means mostly get data

(iii) Highlighting :-

→ we use colouring for highlighting

→ colouring is based on system or candidate's interest.

Miscellaneous Capabilities:-

There are 4 types of miscellaneous capabilities. They are

i) Vocabulary (iii) Canned query
(ii) Iterative search (iv) Z39.50

(i) Vocabulary :-

→ It considers the searched word along with the relevant words in search history along with their occurances to compare among themselves.

Ex:- compromise
Comp
Computer
compulsion
compulsive
compulsory

Y → eliminated by taking in search history

u t
u l s i
u l s i
alphabetical order

→ It compares every u l s s letters in every words in alphabetical order.

→ It gives relevant search based on alphabetical order.

(ii) Iterative search:-

→ Iterative means 'loop'.

→ It searches until we get the required information.

→ It is same as "for loop" as in for loop it get iterated until the condition is satisfied.

→ Till we get required info we search by making changes.

(iii) Canned Query:-

→ Searched from database

→ when we search in IRS we get relevant info.

→ First we process the query then we apply this query to all.

→ Canned Query Cannot be used to all systems

→ It is used for one system & one database.

(iv) Z39.50:-

In 1991,

→ ISO standards they developed which can be used for multiple systems and to access multiple data at a time & for multiple databases

→ latest version is 1995.

Digital Library:-

→ It stores the information (books, material) in digital form, to access & retrieve easily.

→ It is a collection of digital objects that can include text, visual material, audio, video materials stored as electronic media, in formats (JPEG, mp3 etc) along with for accessing, organizing and retrieving the contents.

→ Data source: books, multimedia

→ Converts hard copy to soft copy.

→ Can be accessed (i.e. data) by everyone.

Uses/Advantages:-

→ No need to search the books.

→ search is easy

→ Can be searched through author name, title.

Disadvantages:-

→ Can be worked only through network.

→ It requires devices like phones, pcs.

→ so, in few places there might be non-availability of signal to access the information.

Data warehouse

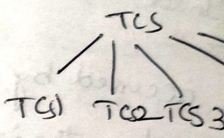
→ It provides security

→ It has more st

→ we can store of data.

→ we can get the

Ex:- TCS has 5 b Can store all b in data warehous get the informat



→ It cannot be a everyone.

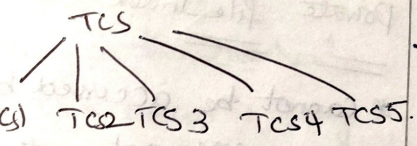
→ It is accessed people

→ Authorised people access the data wherever they w

Data warehouse :-

→ It provides security of the data
→ It has more storage capacity
→ we can store large amount of data.
→ we can get the data easily.

Ex:- TCS has 5 branches. It can store all branches info. in data warehouse. It can get the information easily.



→ It cannot be accessed by everyone.
→ It is accessed by authorized people
→ Authorized people can access the data whenever & wherever they want.

Unit - 2 :-

Cataloging & Indexing :-

→ Both cataloging & Indexing are same
→ Cataloging :-
It is the process of systematically recording & organising information about materials such as books.
→ Multimedia can be of images, audio, video.
→ To perform cataloging or indexing we format the multimedia to text format i.e., Document.
→ Cataloging is performed in "Alphabetical order".
→ we perform indexing on the data in document.

Ex:- The difference b/w cataloging & indexing is

Catalogue - books related to CSE in library
Index → list of topics in each book.

→ we can perform cataloging by author, title, subject name

Types of Catalogue :-

1) Author catalogue :-
→ Cataloging through author name.

It can be producer name, writer name, translator name or publisher name

Ex: RK Narayan ⇒ Author name.

2. Title Catalogue :-

→ Cataloging through title

Ex: Title ⇒ IRs material

③ Author & Title Catalogue :-

→ Cataloging through author name & title.

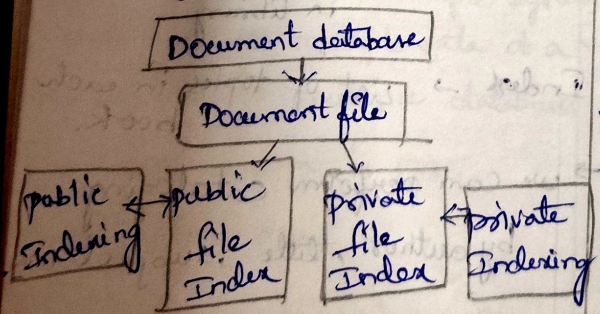
④ subject Catalogue :-

→ Cataloging through/by title of publications.

→ mostly we will use publications for cataloging in this case.

Indexing :-

Indexing can be manual or automatic.



→ manual indexing will be done by us by taking more time

→ Automatic indexing is done automatically by system by taking less time.

Public file Index :-

→ can be accessed by everyone

→ more file storage capacity

→ It has less files

→ Can perform manual or automatic indexing

Private file Index :-

* Cannot be accessed by everyone. only authorized people can access.

→ It has more files.

→ It has less storage capacity

Goals / objectives of Indexing :-

→ It can reduce user's overhead.

→ If the searching is online then relevant data & if the searching is offline then exact data.

Multimedia :-

whatsapp - chat - Export - mail
↓ Drive
- link
- video
- Audio
- emojis
↓ saved as zip file

Automatic I

The system perf
document wi
→ we can use
neural
have diff.
different com
→ Based on
algorithms
time of inde
→ It reduces
→ selection
→ Indexing can
and unweight
methods of I

① weight :-
→ we take li
→ based on
of word.

② unweight
→ It does no
document

IRs mate
Index
Catalogue

dexing is done
by system by

x:-
ed by everyone
orage capacity.
iles
nanual or automatic

s:-
cessed by
ly authorized
ers.
files.
storage capacity

of Indexing:-
user's

hing is online
data & if the
fline then

- Export - mail
Drive
↓
saved
as
zipfile

Automatic Indexing:-

The system performs indexing on document within few seconds.

→ we can use ML, NL, CNN, ~~vector~~ neural complexity & these have diff. algorithms with different complexities.

→ Based on complexities of algorithms we have quality & time of indexing.

→ It reduces the overhead.

→ selection process is easy.

→ Indexing can be done by weight and unweight.

methods of Indexing:-

① weight :-

→ we take link document

→ based on frequency of occurances of word.

② unweight :-

→ It does not take link document.

IRS material
Index
Catalogue

Vocabulary:-

→ It is different for different languages.

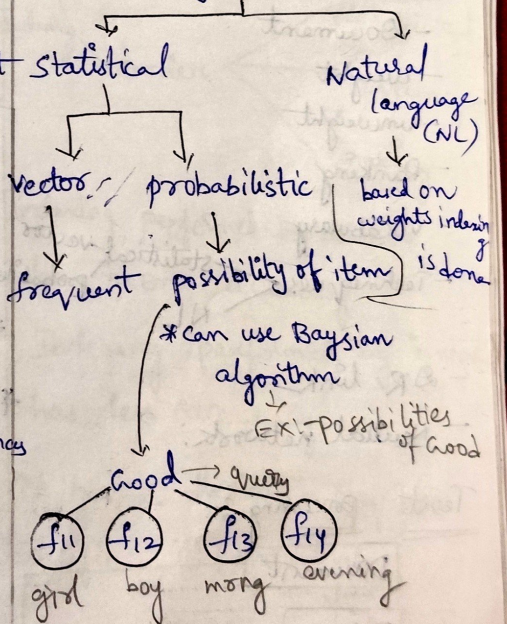
→ There are 2 types, They are author, indexing.

→ Can perform manual indexing.

Techniques of Indexing:-

1. statistical
2. Natural Language (NL)

Indexing Techniques



Ranking:-

$\in 3 - C2 \rightarrow 10$

$\in 3 - C4 \rightarrow 15$

$\in 3 - C1 \rightarrow 3$

so, the ranking is, $\in 3 - C4 \rightarrow 15$, $C3 - C2 \rightarrow 10$, $\in 3 - C1 \rightarrow 3$

DR-line (Document Retrieval line)

→ It doesn't require weighted, unweighted, statistical, natural language .....

→ It requires type of document, set of concepts, set of items in set of concepts.

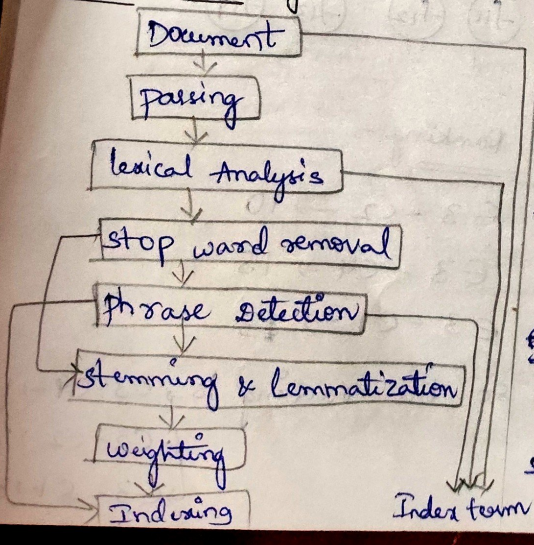
Neural Network :-

→ It is based on words & synonyms.

Automatic Indexing Topics :-

- Document
- weight
- unweight
- Ranking
- Vocabulary
- Techniques < statistical (vector, probabilistic), NL
- DR link
- Neutral networks.

Text processing :-



Document :-

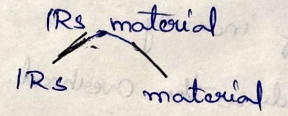
→ It is a full text processing

→ It can be of any language

→ If we use different language then we have different (language) vocabulary.

Passing :-

A single large document is divided into parts.



Lexical analysis :-

→ we will apply the lexical analysis to the document which was divided.

Ex :- Here IRS contains 3 words

* Text = Information Retrieval system.
=> Tokens = Words
* after Tokenization = ["Information", "Retrieval", "system"]
=> Tokenization = we can no longer divide the words (or) (spliting text into words or terms)
=> Tokenisation is performed in lexical analysis.

stop ward removal :-

→ Transforms the plural sentences/words to singular sentences/words

Ex: Search engines are the most visible IR applications [:. plural]
↓
search engine most visible IR application [:. singular]

stemming :-

→ Even though it will generate the

Ex: search eng
most visible
↓
automatically
↓
Search engine
visible informat
application.

→ Do not follow

* stemming & Le

→ we need to ta
with basic

→ Lemmatization
basic structure

→ A word is con
basic structure

Ex :- V1 ← V2
Child ← Ch
↓
basic
structure

Weighting :- It assig
to words in a document
weights (like 1 for pres
0 for absence) based
significant they are

Indexing :-

After weighting, we
perform indexing.

stemming :-

→ Even though we miss a letter it will automatically generate the corresponding word.

Ex:- search engine are the most visible IR applic
↓
automatically
↓
search engine are the most visible information retrieval application.

→ Do not follow dictionary.

* stemming & Lemmatization :-

→ we need to take words with basic structure.

→ Lemmatization requires basic structure.

→ A word is converted to basic structure in lemmatization.

Ex:- $V_1 \leftarrow V_2, V_3, V_4$
Child ← Children
↓ ↓
basic structure — word

Weighting :- It assigns importance to words in a document by giving weights (like 1 for presence and 0 for absence) based on how significant they are.

Indexing :-

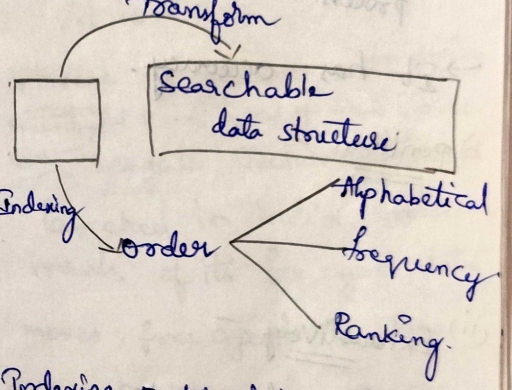
After weighting, we will perform indexing.

Phrase Detection :-

→ count the no of nouns in the sentences.

Ex:- search engines are the most visible information retrival applications } 2
↓
2 nouns are there in this sentence.

Scope of indexing :-



→ Indexing performed by author has more accuracy

→ Indexing performed by index has less accuracy.

→ It has 2 factors. They are
i) Specific ii) Exhaustively

factors — Specific — processor — Intel, pentium
Exhaustively

i) Specific :-

→ It has processor for indexing & it performs indexing which gives less relevant information.

Processor :-

→ It provides relevant info.

→ The relevant info contains types, mini, pc.

→ It has more content.

→ 2 types :- * Intel , * pentium.

a) Intel :-

→ It is direct search

→ It is present somewhere in relevant information of process.

→ It has accuracy.

b) pentium :-

(ii) Exhaustively :-

→ on-board concept.

pre ordination :-

→ Before search, the processor will perform indexing

post ordination :-

→ After search, implementation is by "AND" ing index terms

Information Extraction :-

IRS

goal : Relevant Information

I/p :- Query → [ ] ⇒ Identified processing tokens
words

O/p :- unstructured data.

Result type :- Document (or) webpage

Tools :- Google search, Chat GPT.

→ Information ext
transforms unstr
to structured

→ we will check
from texts &
(i.e., names, ne
while transformin
unstructured data
data.

Example for unstr

Tata Tea Ltd will
stake in Energ
to coca-cola Co
$1.2 billion

converted to str

structured data.

Event = sell

seller = Tata tea

Buyer = coca-cola

stake = 30%

sale value = 1.2 V

Components of Inform

1. Lexical & morph
Processing :-

→ performs Tokeni

→ Next, stemming &

→ Identify nouns,

→ This lexical & morpholog
understanding individual

→ Information extraction transforms unstructured data to structured data.

→ we will check facts, events from texts & entities (i.e., names, norms, ----) while transforming from unstructured data to structured data.

Example for unstructured data:-
Tata Tea ltd will sell its 30% stake in Energy brands to coca-cola company for $1.2 billion.

↓

converted to structured data

↓

structured data.

Event = sell
seller = Tata tea ltd.
Buyer = coca-cola company
stake = 30%
sale value = 1.2 value.

Components of Information Extraction.

1. Lexical & morphological processing:-
→ performs Tokenization.
→ Next, stemming & lemmatization
→ Identify nouns, pronouns.
→ This lexical & morphological processing means understanding individual words & their basic form.

2. Syntax:- understanding sentence structure (grammar)
→ basic phrases.
→ verb group relation
→ grammatical relationship

3. Template:-
→ extracted facts
↑ Converting to structured data.
→ identify what structure it is.

Phrase Extraction:- Identifying meaningful words instead of single words.
Ex:- Seagate Technology has launched in India three models of its free agent data moves, free agent Go, free agent Desktop, free agent pro - targetting the consumer segment.
Company = seagate Technology
verb group = has launched
Location = In India.
Product Names = Free agent datamoves, free agent Go, free agent Desktop, free agent pro-targetting the Consumer segment.

single words
1) seagate,
2) Technology,

meaningful words
seagate Technology

1st unit summary :- MT-1 syllabus :-

* Introduction
* Definition
* DBMS & IRS
* Objectives of IRS.
  - Overhead
  - Precision
  - Recall.
* Functional overview
  - Item Normalization
  - Selective Dissimination of Info.
  - Document DB
  - AfB
* Relationship b/w DBMS & IRS
* Search capabilities
  1. Boolean logic
  2. NLP
  3. proximity
  4. Contiguous word
  5. Fuzzy searches
  6. Term masking
  7. Numeric/Date Range
  8. Concept/Thesaurus
* Browse capabilities
  1. Ranking 2. Zoning 3. Highlighting
* Miscellaneous capabilities :-
  1. Vocabulary
  2. Iterative
  3. Canned query
  4. Z39.50.

* Digital library and Data Warehouse

unit-2 :-

Cataloging :-
↳ Types :- 4types.

Indexing :- Automatic, manual.
↳ objectives.
→ scope of Indexing
→ precordination, post ordination

* Automatic Indexing
  → methods
  → Advantages
  → selection
  → frequency & Ranking
  → Vocabulary
  → Techniques
  → DR link
  → Neural network.

* Text processing.

Information Ext
metrics :-

1. precision :-
How accurate
precision = correct
total
by

② Recall :-
How much relevan
is retrieved.
Recall = correct s
Syst
Total co
in the

Ex :- Google is hirin
engineer in Bang
salary is 12LPA
requires python
ML skills.

Actual (correct i
field = value
1. company → google
2. Job title → s/w
3. Location → Bang
4. salary → 12LPA
5. skills → python

System :-
field = value
1. company → Google
2. Job title → s/w
3. Location → Bangal
4. salary → 12LPA
5. skills → python
6. Experience → 2

Information Extraction metrics:-

1. precision :-

How accurate the system

$$\text{precision} = \frac{\text{correct slots filled by system}}{\text{total slots filled by system}}$$

② Recall :-

How much relevant information is retrieved.

$$\text{Recall} = \frac{\text{correct slots filled by system}}{\text{Total correct slots in the text.}}$$

Ex:- Google is hiring a s/w engineer in Bangalore. The salary is 12LPA. The job requires python, SQL and ML skills.

Actual (correct information) :-

field = value (5 correct)

1. company → google
2. JOb title → s/w engineer
3. Location → Bangalore
4. salary → 12LPA
5. skills → python, SQL, ML

System :-

field = value

1. company → Google ✓
2. JOb title → s/w engineer ✓
3. Location → Bangalore, Hyderabad
4. salary → 12LPA ✓
5. skills → python, SQL, Java
6. Experience → 2 years

* 3 correct in system.

precision = 3/6 = 0.5 = 50%.

Recall = 3/5 = 0.6 = 60%.

Ex:- Tata tea limited will sell its 30% stock in energy brands to coca cola company for $1.2 billion.

Actual :-

1. Event → sell
2. seller → Tata tea limited
3. Buyer → coca cola company
4. stocks → 30%.
5. sale rate → 1.2 billion
6. Brand → Energy brand.

System :-

1. server → Tata tea limited ✓
2. Buyer → coca cola ✓
3. stocks → 30%. ✓
4. sale Rate → 1.2 billion ✓ (5 correct)
5. brand → Energy brand ✓
6. year → 2020 (this is extra field)

Recall = 5/6 = 0.833

precision = 5/6 = 0.833

∴ Recall = precision

Note :-

we can have conditions :-

P > R, P = R, P < R where P = precision, R = Recall

F-measure :-

This is a factor to calculate the relation b/w precision and recall

$$F = \frac{(\beta^2+1).P.R}{(\beta^2.P)+R}$$

β = weight parameter

β values :-

β values in Factor F can be

$\beta = 1$ → equal importance

$\beta > 1$ → more weight to Recall

$\beta < 1$ → more weight to precision

Ex :-
for previous problem,

As Recall = precision so they have equal importance.

∴ we take $\beta = 1$

$$\Rightarrow f = \frac{(1^2+1).\frac{5}{6}\times\frac{5}{6}}{(1^2.\frac{5}{6})+\frac{5}{6}} = 0.933$$

Data structures :- (DS)

→ Collection of data

→ To store data in structured format.

Ex :- stack, Array, Queue, linked list.

→ It can be used in C, C++, Java

→ Data structures has less storage capacity.

→ WKT, IRs has unstructured data & DS is used to store structured data. so, we use concepts like

- stemming
- Inverted file
- PAT
- N-Gram
- Hyper text & XML
- signature file

→ RGUKT

Ex :- E3 - CSE - C2

Branch, section, 2210942

stemming :-

→ converting a word to its root word.

→ root word is called as stem

→ IRs efficiency is improved.

→ Improved recall

Ex :- word - similar meaning

connect
connected → root word (stem) → connect
connecting
connections → we store the root word i.e., connect.
connection

⇒ we can decrease the usage of storage capacity by storing only the root word instead of all similar words.

→ we can easily data.

Disadvantages :-

→ when the word meanings and store root word we are retrievi it gives diffe leads to time process.

→ stemming can on the words with 'I, A, R, E'

→ stemming can b using 4 Algorith

① Table look up
② Porter Algorith
③ Dictionary looku
④ successor stemm

Table lookup :-

→ Index terms ar in table and sea the required value

→ Index is used to information easily a less time.

→ Each index is sto table in its root

→ Required word is th not in table for

has less storage

unstructured
used to store
so, we use

XML

942

to its

d as stem
improved.

meaning
word (stem)
connect
ore the
word i.e.,
connect.
use of
storage
only the
all

→ we can easily retrieve data.

Disadvantages:-

→ when the word have diff. meanings and but we store root word then when we are retrieving the word it gives different data and leads to time taking process.

→ stemming cannot be applied on the words starting with 'I', 'A', 'R', 'E'.

→ stemming can be applied using 4 Algorithms.

① Table look up
② Porter Algorithm
③ Dictionary lookup
④ successor stemming.

Table lookup:-

→ Index terms are stored in table and search for the required value.

→ Index is used to get the information easily and takes less time.

→ Each index is stored in table in its root form

→ Required word is there or not in table for that work is done.

Advantages:-

→ Easy search of information

Disadvantages:-

→ semantic information
→ storage (duplicates can be executed)
* overlapping

② porter stemming Algorithm

→ Depends on set condition
→ Forms a stem from the word
→ words formed by letters.
→ consonant
vowels → a, e, i, o, u

$(CVC)^m$ ⇒ format

m ⇒ power means how many times VC repeats.

Ex:- T R O U B L E
C C V V C C V
C V C V
m = 1

⇒ end with X ⇒ X stem (Ex:- xerox)

⇒ *V* ⇒ start & end with vowels. Ex:- Apple.

⇒ V stem → end with vowel (CV stem)

⇒ C-stem → end with consonant

⇒ d stem → end with double consonant

→ O ⇒ CV sequence ⇒ O-stem
↓
w, x, y ⇒ do not consider.
Ex:- xe (not a O stem)

Suffix	Replacement	Example.
Sses	ss	stresses → stress
ing	Null	making → mak
y	i	Happy → Happi
ated	at	Inflated → Inflat
aliti	al	formaliti → formal
Icate	ic	Duplicate → Duplic
Able	Null	Adjustable → Adjust
e	Null	Inflate → Inflat

STRESSES
C C C V C C V C
C O C O
∴ m=0

FORMALITI
C V C C V C V C V
C I I V
O
⇒ m=2

$C(VC)^2V$

Successors :-

length of prefix

→ smallest unit → smallest word

Takes smallest word from smallest unit.

→ It has 4 methods

(i) cut off method
(ii) Peak & plateau
(iii) Complete wood
(iv) Entropy

(i) cut off method :-

* Based on stem length

word → READABLE

Ex :- Corpus :- ABLE, ABILITY, RIDE, ROPE, READ, READS, READING, READABLE

Word	successor	Letter.
R	3 (I, O, E)	E, I, O
RE	1	A
REA	1	D
READ	3	S, I, A
READA	1	B
READAB	1	L
READABL	1	E
READABLE	Null	-

(ii) peak & plateau :-

* whenever we are dividing into 2 parts - 1st part length < 2nd part length.

* It doesn't need to be meaningful.

(iii) complete word :-

* It needs to be meaningful when we are dividing word.

Ex :- READ / ABLE

(iv) Entropy :-

* probability

formula :- $H = \Sigma p(j$

=> to calculate proba

Actual Formula :- H

where,

Dak = word length

Ex :-

Reading — Prefix :
Reader — Dak = {Re
Readable — Readab
Read — Dak : 4

Successor's of Read

Reading → i, Reader
Readable → a, Read

$Dak_j = \frac{|\text{successor } j|}{|Dak|}$

i → 1, e → 1, a → 1, y

$P(j) = \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$
P(i) P(e) P(a) P(y

$H = -\Sigma p(i) \log_2 P(j$

$= -\left[\frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{1}{4}\log\right.$

$\frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{1}{4}\log$

$= -4 \times \frac{1}{4}\log_2\left(\frac{1}{4}\right)$

length

DABLE

E, ABILITY,
READ, READS,
DABLE

	Letter
	E, I, O
	A
	D
	S, I, A
	B
	L
	E
	-

dividing
part length

be

ningful
g word.

(iv) Entropy :-

* probability

formula :- $H = \Sigma p(j) \cdot \log_2 p(j)$

=> to calculate probability

Actual Formula :- $H = \frac{|Dak_j|}{|Dak|}$

where,

Dak = word length, $Dak_j = j$

Ex1:-

Reading
Reader
Readable
Read

Prefix :- Read

Dak = {Reading, Reader, Readable, Read}

Dak : 4

successor's of Read

Reading → i, Reader → e,
Readable → a, Ready → y

$Dak_j = \frac{|\text{successor } j|}{|Dak|}$

i → 1, e → 1, a → 1, y → 1

$P(j) = \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}$

$P(i)\ P(e)\ P(a)\ P(y)$

$H = -\Sigma p(i) \log_2 p(j)$

$= -\left[\frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{1}{4}\log_2\left(\frac{1}{4}\right)\right]$

$= -4 \times \frac{1}{4}\log_2\left(\frac{1}{4}\right)$

$= -\log_2 2^{-2} = -(-2) \times 1$

$= 2$

$\therefore H = 2$

Ex:-

Playing → i = 1 ⇒ $\frac{1}{4}$ Prefix: play

Player → e = 1 ⇒ $\frac{1}{4}$ Dak = {playing, player, played, plays}

played → e = 1 ⇒ $\frac{1}{4}$

plays → y = 1 ⇒ $\frac{1}{4}$

∴ Dak = 4

$Dak_j = \frac{|\text{successor } j|}{|Dak|}$

$P(i) = \frac{1}{4}$; $P(e) = \frac{2}{4}$; $P(y) = \frac{1}{4}$

$H = -\Sigma p(i) \cdot \log_2 P(j)$

$= -\left[\frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{2}{4}\log_2\left(\frac{2}{4}\right) + \frac{1}{4}\log_2\left(\frac{1}{4}\right)\right]$

$H = 1.5$

Inverted file structure :-

→ document file is taken as inverted list, dictionary, inverted file.

Inverted list => based on methodology we store words full text search

Dictionary - lookup.

Inverted file - Easy to search

Document	(Inverted list)	Dictionary	Inverted file
DOC#1			
computer, bit, byte	computer (2)	computer	computer → 1,3,4
DOC#2	bit (2)	bit	bit → 1,3
memory, byte	byte (3)	byte	byte → 1,2,4
DOC#3	memory (2)	memory	memory → 2,3.
computer, bit, memory			
DOC#4			
byte, computer			

Ex:- bit 1 (10, 12, 18)

↑ positions in inverted file.

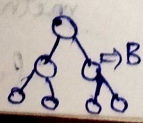
Search bit	10	1	
	12	0	1 → present
	18	0	0 → Not present

→ Inverted file consists of words, numbers & symbols

→ Numbers & symbols are stored based on their priority order.

→ It always consists of current date, month, year.

→ We use B-tree concept (Binary-tree)



→ B-tree

N-gram Data Structure:

→ we can divide the word from 1 to n grams.

word → 1 ..... n grams

1 gram ⇒ word

Bi gram ⇒ wo or rd

tri gram ⇒ wor ord

→ we don't use inter word symbols in 1 & Bi grams

→ Bi gram is used to include words.

→ In tri gram sometimes we use inter word symbol & sometimes we don't

→ Inter word symbols include #, blanks, semicolon, colon

→ Inter word symbols can be used as suffixes.

wo # or

→ Tri gram or N-gram or Penta-gram are used to identify the mistakes & make them correct & meaningful.

→ "Damerau" decided the categories of errors.

S.NO	Error Category
1.	single char Insertion
2.	single char deletion
3.	single char substitution
4.	Transposition of 2 adj chars

→ "Zamora" corrected spellings, errors

Advantages:-

Σ = 5

Σ = {a, b, c, d, e}

n = 1-gram = $(5)^1$ =

= bi gram = $(5)^2$ = 2

= tri gram = $(5)^3$ = 12

Σ n gram = 5^n

∴ max segmen

n = n-gram

Disadvantages:-

* Uses semantic con

i.e., precision ↑ so irrelevant data.

S.NO	Error Category	Example
1.	Single char Insertion	commputer
2.	single char deletion	Coputer
3.	single char substitution	compiter
4.	Transposition of 2 adj chars	computter

→ "Zamora" corrected the spellings, errors.

Advantages :-

$\Sigma = 5$

$\Sigma = \{a, b, c, d, e\}$

n = 1-gram = $(5)^1 = 5$

= bi gram = $(5)^2 = 25$

= tri gram = $(5)^3 = 125$

Σ n gram = 5^n

∴ max segment = Σ^n

n = n-gram.

Disadvantages :-

* Uses semantic concept i.e., precision ↑ so more irrelevant data.

PAT Data structure :-

→ practical algorithm to retrieve information.

→ coded in alphanumeric

Bit → 0 & 1

→ storing searching

→ Binary tree compress

→ we take/consider suffix & substring to construct PAT tree.

→ word is transformed to coded form (i.e, 0 & 1) & we store it.

→ After storing the coded form we can search.

word → code (0 & 1) → store → search.

→ B-tree starts from prefix.

→ while constructuring PAT tree using substring we need to follow particular order.

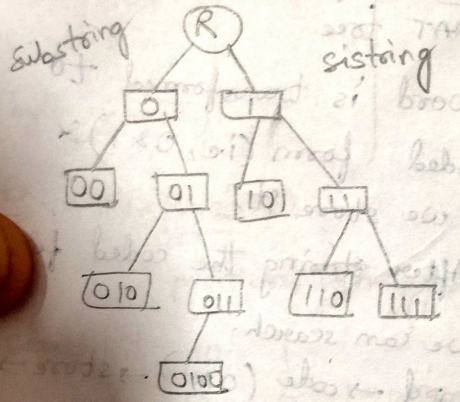
C
Com
Comp
Compu
Compute
Computer

} possible substrings of computer.

1 - computer
2 - omputer
3 - mputer
4 - puter
5 - uter
6 - ter
7 - er
8 - r

} sistring of computer.

→ There is no need of being in same level in PAT tree.



sistring
substring

Ex:- 100111001

sistring 1 → 100111001

sistring 2 → 00111001

Ex:- Economics for warsaw is complex.

* we also consider spaces.

sistring 1 : economics for warsaw is complex.

2 : nomics for warsaw is complex.

signature file data structure

→ converts word into coded format (i.e., 0's & 1's)

→ If the text / document / word has more than 2 words we need to do partition & then convert it into coded format of 0's & 1's which is of 16 bit using "OR" operator. This is called signature file data structure.

Ex:- computer science graduate students study.

words.

computer → 0001 0110 0000 0110

science → 1001 0000 1110 0000

graduate → 1000 0101 0100 0010

students → 0000 0111 1000 0100

study → 0000 0110 0110 0100

Apply OR operation perform calculations

1001 0111 1110 0110

→ we store the signature file in data structure.

→ we search through the coded format which is stored

HTML & XML :-

→ HTML : Hypertext markup languages.

→ we use HTM webpages.

→ when we searc webpages link which contain information.

→ webpages inter existed documen contain relevant related to se

→ The main role is "inter linkin

→ XML is used to transport.

Hidden Markov

→ It is a stati probability m

→ system {output

→ we need to see state in obse

Ex:- word :- match Score.

Observation :- matc

by observation it to spo

∴ Relevant informa hidden markov

Data structures

into coded

o's & i's)

ocument / word

2 words we

tition & then

coded format

ch is of

" operator.

signature

re.

e gradiate

0000 0110

110 0000

100 0010

00 0100

0 0100

0110

Apply OR operation perform

Calculation

ere file

the

is stored.

markup

→ we use HTML to create webpages.

→ when we search in webpages, webpages link to the files which contain relevant information.

→ webpages interlinks with the existed documents which contain relevant information related to search.

→ The main role of HTML is "inter linking"

→ XML is used to store & transport.

Hidden Markov model :-

→ It is a statistical probability model.

→ system {output / observation}

→ We need to see hidden state in observation

Ex:-

word :- match, player, score.

Observation :- match, player, score.

by observation it is related to sports.

∴ Relevant information = hidden markov = "sports".

Hidden (S) ⇒ S_1 = sports, S_2 = politics

Observation : cricket, election

Components :-

1. Initial probability (Π) :-

Sentence → state

4 - sentences, 3 - sports, 1 - politics

Π (state) = $\frac{\text{state start count}}{\text{Total sentences.}}$

sports = 3/4

politics = 1/4

Transition probability (A) :-

$A(i-j) = \frac{\text{transition count from } i \text{ to } j}{\text{Total transition from state } (i-j)}$

sports → sports → politics → sports

sports → sports = 1

sports → politics = 1

politics → sports = 1

Emission probability (B) :

Sports	words
goal, match, play	

P(goal / sports) = 1/3

P(match / sports) = 1/3

P(play / sports) = 1/3.

Unit - 3 :-

Automatic Indexing :-

→ There are 2 types of Indexing

Indexing
- Manual
- Automatic

→ Automatic Indexing is performed by system i.e.) - processor

→ based on the system complexity we perform the automatic indexing

→ multimedia files are converted to text & next to document to perform indexing.

→ Indexing is done by :
- Rank
- frequency
- weight
- unweight

- weight :-
  we use link document to search
- unweight :-
  we don't use link document to search.

Techniques of Automatic Indexing :-

→ There are 4 types of techniques

(i) statistical :-
- Vector
  based on frequency
- probabilistic
  * usage of bayesian N/w
  * Calculated based on rank & freq.

(ii) Natural Language processing (NLP)
- weight based indexing.
- we can take information extraction (IE).

(iii) concept

(iv) Hypertext

(i) statistical :-
- Vector
  * based on frequency
- probabilistic :-
  * usage of bayesian N/w.
  * calculated based on rank & freq.

→ It has 7 models.

a. Probability weighting
b. Vector
c. simple term frequency
d. Inverse document
e. Signal weight
f. discrimination values
g. Bayesian model.

frequency
- NO. of occurances of related word

Document frequency :-
- * NO. of documents

containing
Ex: suppose
documents
documents
word.
; documen
Term freque

a). probability
→ As we cann
probability di
probability
* we store th
* Results are i
* Conclusion is
through evidan
* Range
* Frequency
b. Vector weight
* Calculated ba
* It can be calle
statistical
* It uses system
* It contains one
of set values &
representation.

containing related word.

Ex: suppose there are 10 documents but only 5 documents contain related word.

∴ document frequency = 5

Term frequency :-

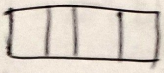
a). probability weighting :-

→ As we cannot calculate probability directly, we use probability algorithms.

* we store the results seperately
* Results are integrated.
* Conclusion is calculated through evidance.
* Range } Calculated based
* Frequency } on this two.

b. Vector weighting :-

* Calculated based on frequency
* It can be called Investigated statistical model.
* It uses system based.
* It contains one dimension of set values & it is fixed representation.



* we also take domain & token values.
* we can also use binary approach.

Ex:- suppose ; signature file is stored in data structure and here the word can be in coded format. so, there is a need to use binary approach.

c. simple term frequency :-

* weight - based on frequency, searching is done
* unweight - directly searching is done without frequency
  - It displays the pg.no where the word is present

* For weight, it gives result based on already calculated values.

d. Inverse document :-

* Rank & frequency are calculated inverse.

Ex:-
One with high marks gets last rank
" " low " " high "
→ Here, priority changes

e. signal weight :-

* It uses frequency distribution
* The word along with frequency is stored in list (or) table. before indexing.
* And based on priority, the indexing is performed.

f. Discrimination values :-

* All items are represented as same.
* Document contains no. of items and are represented as same which are unique in identity & then we calculate value.

(i) NLP :-

* Item indexing is done by statistical & semantic concept.
* statistical concept - uses probability
* semantic concept - based on meaning
  Ex :- Doctor
* Thematic concept :- based on main concept
* The processing of NLP it results in phrase.
* Based on phrase, indexing is performed.

* Text processing is used.

full Text
↓
parsing
↓
lexical analysis
↓
Stop word analysis
↓
phrase detection
↓
stemming & lemmatization
↓
weighting
↓
Indexing

* we don't use lexical analysis and stemming & lemmatization in NLP.
* Directly from phrase detection, indexing is done.
* words are seperated
* promixity : distance b/w 2 words.
* It considers frequency of occurances.
* syntactical model : The structure of the text/word.
* Based on syntactical model, indexing is performed.

phrase generation :-

semantic concept :-

* we take text
* Term based calculation Performed.

Thematic concept :-

* we take text
* Term based calcul
* Performed.

statistical concept :-

* we considers/takes
* It is easy.

cohesion(kh) = size fact
↓ (Pair
for ↓
term K
base
Total F_k

→ cohesion is calcul
find strength b/w

→ proximity restric
distance b/w & w
⇒ $w_1 - w_2$

⇒ side factor ⇒ voc

Ex :- If we take wo
player, match, go
belongs to

⇒ so here we find
b/w player & spo
match & sports o
sports.

Ex :- Statistical form
Industrious, Int

Thematic concept :-

* we take text
* Term based calculation performed.

statistical concept :-

* we considers/takes one word.
* It is easy.

cohesion(kh) = Size.factor × (Pair-frequency) / (Total F_k × Total f_h)

(for term base; Pair-frequency: k, h)

→ cohesion is calculated to find strength b/w 2 words.

→ proximity restrict the distance b/w 2 words.

⇒ $w_1 - w_2$

⇒ side factor ⇒ vocabulary.

Ex :- If we take words like player, match, goal belongs to "sports"

⇒ so here we find strength b/w player & sports or match & sports or goal & sports.

Ex :- Statistical form :-

Industrious, Intelligent student.

NLP :- (we can write in so many forms)

Industrious student

Intelligent student

Industrious Intelligent student

⇒ parts of speech ⇒ checks

⇒ Do lexical analysis.

Lexical analysis :-

TOkenization of word or sentence takes place.

→ lexical is parts of speech → adj, nouns.

⇒ TTP ⇒ Tagged Text parsing
↳ has 400 above grammar rules.

→ class TTL ⇒ 400 grammar rules

→ determiners → a, the

→ singular noun ⇒ paper, structure

→ plural noun ⇒ operation, data

→ preposition → in, by, of

→ adjectives ⇒ high

~~⇒ preposition~~ x

→ present tense verb → present, associates.

⇒ for these words by apply TTP ⇒ parse tree is made and by that tree o/p we do indexing.

Header modifier & Information contribution :-

→ gives word + extra description

→ Advantage : is that if we type the word for search the automatically gives the description of it.

Ex:- grass field, magnetic field } if searched gives description as well with word.

→ Info contribution ⇒ gives probability that, how many letters a word contains

$(x, y) = w_1 - w_2$

w_1 ↓ how many letters in w_1

w_2 ↓ how many letters in w_2

Concept :-

→ It has document & vector model.

→ we take only necessary words from the corpus from cutoff method so this concept ensures abstraction.

→ High level indexing takes place.

Ex:- word = Automobile

corpus = { Vehicle, transportation, mechanical device, fuel, environment.

→ we use Neural N/W & N... to calculate. like :-

Vehicle ⇒ 0.65

Transportation ⇒ 0.60

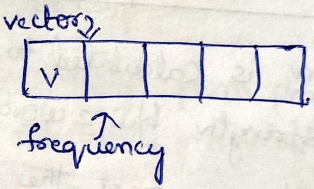
Environment ⇒ 0.35

fuel ⇒ 0.83

mechanical device ⇒ 0.15

Vector model :-

It is a 1D model which takes only necessary words and store based on frequency word.



Hyper text linkages :-

⇒ It links files or websites when searched with words.

→ The order of files will be based on rank.

→ Highest rank will be alloted to the files which has high feedback, views, downloads (most searched document).

Two documents

original ↓ copy → for reference

Issue & ...

→ For ...

→ Automatic ...

→ web spide...

clustering :-

→ Divided ...

→ It is a p... used to ... information

Thesaurus ...

level 3 ↓ ...

* In cluster... similar ... cluster/...

* Relation ... is also di... (word &...

Ex:- Author ...

word

CSE

Issue & mechanism :-

→ For manual Indexing ⇒ yahoo.com

→ Automatic Indexing ⇒ lycos.com, Altervista, google.com.

→ web spider.

clustering :-

→ Divided into clusters.

→ It is a process which is used to find where the information is located.

Thesaurus → latin word.

↓ level 3

↓ It is a dictionary
- has word → meaning
- synonyms
- Antonyms

* In clustering, we divide similar Info into one cluster / group.

* Relation b/w 2 words also is also discussed here (word & Document of words).

Ex:- Author ← publisher, writer.

word — relationship

C1 E3

C1, C2, C3, C4 } Document of words

Domain concept :-

→ similar words into doc/objects

↳ into class → into cluster

into domain ← (has same class objects) (same cluster)

→ Domain has attributes.

→ Relationship ⇒ title - author

⇒ class rule → semantic, concept

→ obj size < class (should be always less.)

→ obj does not dominate the class.

Computer
↓
microprocessor
/ \
Intel pentium

* we select 2 words whose relationship is max which is came by applying class algorithm.

Ex:- If in computer → Intel has 90% then (that most relevent one) that will be taken.

Thesaurus Generation :-

① word Coordination approach :-

* phrases + individual words are taken.
* Each phrase is compared / checked with all individual words and checks whether that pair (phrase + individual word) forms as a cluster or not.

② word Relationship :-

3 types
① Equivalence
② Hierarchical
③ Non Hierarchical

① Equivalence :-
Same meaning

② Hierarchical :-
Term + specific entries of that word are taken

Ex:- Computer (class name)
/ \
microprocessor pentium

③ Non Hierarchical :-
Objects + attributes are taken
or entities

Ex:- Employee → ID, name
↓ attributes
Obj

③ Homograph resolution :-
→ words with different meanings are taken.
→ more than one word is taken.

Ex:- ① grass field
magnetic field
strength b/w 2 words is calculated and most used word is taken.

② Reading books → most used
Reading Novels so select this.

④ Vocabulary :-
→ Normalisation is done
→ stem or complete word is taken based on vocabulary (Parts of speech) for results.

Generation methods :-
① hand - crafted.
② CO - occurence
③ header - modifier.

Manual :-
① KWOC → keyword out of context
② KWIC → keyword In Context
③ KWAC → keyword and context

① KWOC :-
Keyword out of

Ex:-

Term	freq	Item
Chips	2	doc2
computer	3	doc1
design	1	doc
memory	3	doc3

② KWIC :-
→ keyword In contex
→ we take keywo
also its contex
⇒ we put "/" symb
and word.

Ex:- Term/Text :- com
contains

Keyword	
Chips	computer contains
computer	design co Chips
design	contains Chips/ co
contains	memory desig
memory	Chips/ co Contain

① KWOC :-

Keyword out of context.

Ex:-

Term	freq	Item Ids.
chips	2	doc2, doc4
computer	3	doc1, doc4, doc5
design	1	doc4
memory	3	doc3, doc4, doc5.

② KWIC :-

→ Keyword In context.

→ we take keyword and also its context.

⇒ we put "/" symbol after and word.

Ex:- Term/Text :- computer design contains memory chips

Keyword	context
chips	computer design contains memory /
computer	design contains memory chips /
design	contains memory chips / computer
contains	memory chips / computer design.
memory	chips / computer design contains

③. KWAC :-

we consider keyword + full Text.

⇒ In full Text we highlight the keyword

Ex:- Text :- computer design contains memory chips

Keyword	Text
chips	computer design contains memory chips
computer	computer design contains memory chips

Term - clustering :-

Term is a word.

→ It will be in vector matrix means combination of rows and columns. ⇒ i.e., < items

Q :- DOC3 = memory, chips < Terms
DOC1 = computer, memory
DOC2 = computer, design

Ex:-

Item/Term	computer	memory	design	chips
DOC1	1	1	0	0
DOC2	1	0	1	0
DOC3	0	1	0	1

To find similarity :-

formula :-

$$sim(Term_i, Term_j) = \Sigma(Term_{i,k} * Term_{j,k})$$

→ we take Term - Term matrix to find similarity.

Term-Term matrix formula:-

	T_1	T_2	T_3
T_1	11	12	13
T_2	21	22	23
T_3	31	32	33

where 11 ⇒ Term 1 × Term 1

12 ⇒ Term 1 × Term 2

13 ⇒ Term 1 × Term 3.

TO find 11 ⇒ T_1 T_1

1 × 1 ⇒ 1

1 × 1 ⇒ 1 } 1+1+0 = 2

0 × 0 ⇒ 0

⇒ similarly find 12, 13, 21, ---33.

	T_1	T_2	T_3
T_1	2	1	1
T_2	1	2	0
T_3	1	0	1

→ In above matrix total 9 values are there and in that 2 are zero so avg = $\frac{9}{7}$ = 1.2

→ TO find relationship matrix we take avg value in integer form and compare it with every term in Term-Term matrix.

avg = 1.2 ≅ 1

if T ≥ avg = 0 | P ⇒ 1

(T: Term)

T < avg = 0

Relationship matrix :-

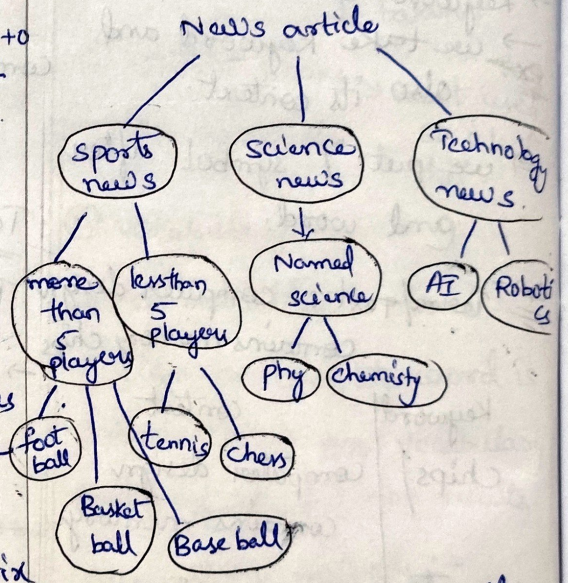
T ≥ 1 = avg

$$RM = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

|V| = |V|

we find similarity by this.

Hierarchy clustering :-

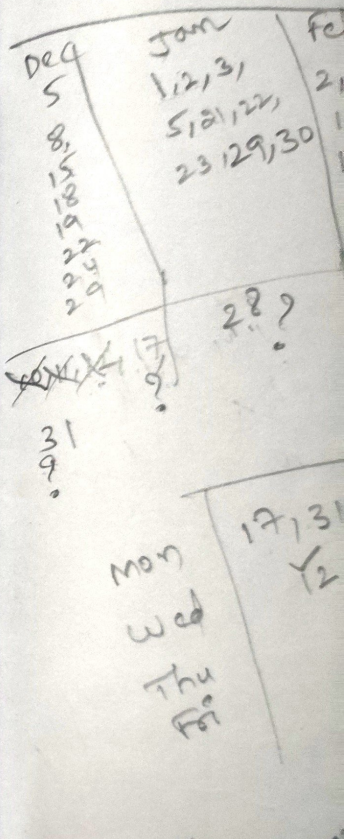


Bottom-up approach :-

* we have to compare/add all children of a parent node and have to store in it and then repeat procss for all nodes.

EX:- Add footBall + Basket ball + base Ball then store value in more than 5 players node. similarly add

tennis & chess to ... value of "less than ... node then ... than 5 players" ... 5 layers" to ... of "sports news"

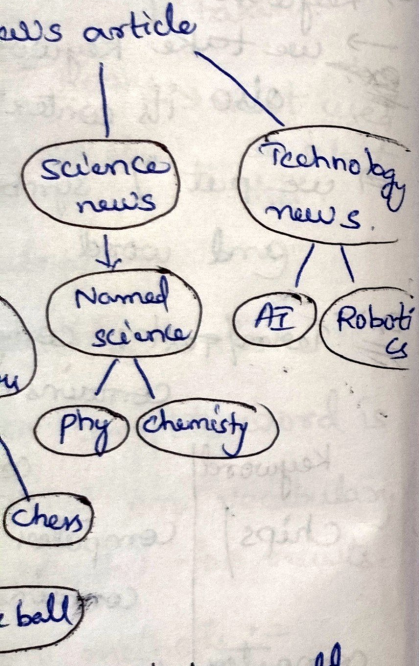


onship matrix :-

1	1	1
1	0	1
0	1	

d similarity by this.

hy clustering :-



p approach :-
e to compare/add all
of a parent node
we to store in it
repeat process for
odes.

football + Basketball +
Ball then store
more than 5 players

tennis & chess to find value of "less than 5 players" node then ~~for~~ add "more than 5 players" & "less than 5 layers" to find value of "sports news".

Dec	Jan	Feb	Mar
5	1,2,3,	2,4,5,	5,9,
8,	5,21,22,	13,16,	11,18,
15	23,29,30	18,23,	20,23,
18		25,	24
19			
22			
29			

Mon	17,	31,	28, 11,
Wed		12,	2, 6, 12
Thu			
Fri			

26th mar.
Attendance ATCD absentees:-
3, 8, 31, 34, 40,
12, 33, 59

IOT attendance
25th Mar
presentees (IOT)
1, 5, 6, 7, 11, 13, 14,
16, 19, 20, 22,
25, 28, 32, 33,
38, 40, 39, 41,
45, 47, 48,
49, 53, 54,
57

Attendance :- 11th Mar
Absentees:-
7, 8, 9, 11, 12, 13, 15,
20, 32, 34, 40, 51,
54,